

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

- A. Decreased prolactin, decreased FSH, decreased LH
- B. Decreased prolactin, increased FSH, increased LH
- C. Increased prolactin, decreased FSH, increased LH
- D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>

Finding Reasoning Paths :

1. **Nipple discharge and hyperprolactinemia**:

The patient presents with spontaneous milky-white nipple discharge , which is a hallmark of hyperprolactinemia . Hyperprolactinemia can result from elevated prolactin secretion due to prolactin-secreting tumors or hypothyroidism .

2. **Hypothyroidism**:

Hypothyroidism is a common cause of hyperprolactinemia , as it suppresses the release of thyrotropin-releasing hormone (TRH) from the hypothalamus , leading to reduced stimulation of the anterior pituitary gland and decreased prolactin secretion .

3. **Hypothalamic dysfunction**:

Hypothalamic tumors (e.g ., craniopharyngiomas) or other hypothalamic disorders (e.g ., empty sella syndrome) can lead to secondary hyperprolactinemia by directly or indirectly affecting the pituitary gland .

4. **Serology and brain MRI**:

The patient's pending serology and MRI results suggest a potential pituitary lesion or systemic disorder that may cause hyperprolactinemia .

5. **Elevated prolactin levels**:

Hyperprolactinemia typically presents with high serum prolactin levels and , if due to a pituitary tumor , may be accompanied by additional symptoms such as amenorrhea , galactorrhea , and infertility .

Reasoning Process :

1. **Clinical Presentation**:

The patient has missing periods , hypolactation , and galactorrhea . These symptoms strongly suggest hyperprolactinemia , a condition characterized by abnormally high levels of prolactin .

2. **Pathophysiology of Hyperprolactinemia**:

- Primary hyperprolactinemia:

This occurs due to elevated prolactin secretion , often caused by prolactin-secreting pituitary tumors (prolactinomas) or other systemic conditions like hypothyroidism or certain medications .

- Secondary hyperprolactinemia:

This occurs when the hypothalamus fails to stimulate the pituitary gland appropriately , leading to reduced prolactin secretion . This can result from hypothalamic tumors (e.g ., craniopharyngiomas), hypothyroidism , or other pituitary disorders .

3. **Differential Diagnosis**:

- **Hypothyroidism**:

This is a leading cause of hyperprolactinemia . The patient's symptoms (nipple discharge , missing periods) could be attributed to hypothyroidism , as hypothyroidism suppresses prolactin secretion and can lead to elevated prolactin levels .

- **Pituitary Tumor (Prolactinoma)**:

This is another common cause of hyperprolactinemia . However , the patient's symptoms (nipple discharge , missing periods) are more consistent with hypothyroidism than a pituitary tumor , as pituitary tumors typically present with galactorrhea without missing periods .

- **Hypothalamic Dysfunction**:

This could explain the missing periods and milk discharge but is less likely given the absence of other symptoms (e.g ., vision changes , headaches).

4. **Serology and Brain MRI**:

- The pending brain MRI may reveal a pituitary tumor or other structural abnormality , but hypothyroidism remains a strong possibility given the patient's symptoms .

- Blood tests for thyroid function (e.g ., TSH , T4) may confirm hypothyroidism , supporting this diagnosis .

5. **Expected Laboratory Findings**:

- **Elevated Prolactin**:

Hyperprolactinemia is defined by elevated prolactin levels .

- **Decreased FSH and LH**:

Hypothyroidism inhibits the secretion of gonadotropin-releasing

1.0

0.5

0.0

-0.5

-1.0